Q. Nos.

* Flow control:- the Flow control measure are taken By the intation of the variable ~~is~~ Sliding window where this window make alot of differnce in a transmission, it helps to control the too much of Data overwhelming at the reciver side and causing collisions.

* where it works upon a simple machism, where for each Data packets the reciever side reciver sends the an acknowledgment that How many space is it still available to store the Data packets.

* according to that acknowledgment count the transmission at the sender's sliding window happen where the sliding window dosnot send a data packets above a given range of the acknowledgment from reciver side.

Example: if a Data at the sendes side is about 5000MBS & the reciver sides sends an Acknowledgement after reciving Data that it can store only about 2000MSS, the sends side dosnot send a Data more than the give range of the reciver side to avoid collision

Congestion control:- This is a protocal or a process used to avoid the congestion and the collision among the Data packets while transmision and its working mechanism is ~~the~~ ~~described~~ described as having the windows at Both the sides.

Q. Nos.

* Where initially a variable Cwind (congestion window at the sender side) & rwind (receiving window at the receiving side).

* The formula on which the mechanism works is

The total data that can be sent = min{Cwind, rwind} //

* Here Starting the congestion window is set to one MSS & we have a limit of Slow Start threshold.

* Where for each acknowledgement sent from the receiving side the congestion window is being exponentially grows its transmission rate

Ex: 1, 2, 4, 8, 16 etc...

* At a point when it reaches the Slow start threshold limit the congestion window is being reset where the SSH (Slow start threshold) limit is equals to half of the Cwind.

$$SSH = \frac{Cwind}{2}$$

&

C = reset to Starting.

if the packets are being missed in between the transmission and the congestion just happens mildly. not to heavy the Cwind is again reset when.

$$SSH = \frac{Cwind}{2}$$

&

$$Cwind = SSH_{11}$$

This protocols are being executed to recovery the packets which is lost.

At the transmission layer the Data arrays are handled at the protocols like. Recongonizing the intial sequence number & calculating the checksum and verifying to the check sum before opening the playhood data at the receiving end to check whether the Data is corrupted or not.

Module-3,,

Q6
b
Answer:-

IPV4 Structure:- It is a Structure that consists of the variable to Headers and are less efficient compared to the IPV6, where due to variable Heady size it Harder to look at the exact playhood data. and it consist of the multiple ~~Comments~~ componets through which the transmission can be done.

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These components are:

Version: 4 bytes

It Specifies the what version of the protocol is being used to conduct the transmission of the IPV4.

Header:
4 bytes:

It consist of the ip source & the destination address and its multiple 5 times to get the original size 5×4=20 bits. and consist of the playhood Data.

protocol:
Size: 8 bits.

This is a Standard Set of instructions used to conduct the transmission.

TTL-Time:
Size: 8 bits:

This is a important component which prevents a Data host in a endless hoop across the network. This Component keep the hop count of the Data packets which is being travelled.

Q. Nos.

Ip Source address:

* It is the addres of the source from which the Data is being transmitted.

Ip destination address:

* It is the addres of the destination where the Data packets have to be transmitted.

Checksum:- the check sum calculation is being done on the bits to prevent the bits from getting corrupted when the Both Sender & receiving side must perform the checksum to retrieve get a error free data if the calculation on the checksum dosn't match the Data ~~packeted~~ packet is thrown and requested for retransmission.

